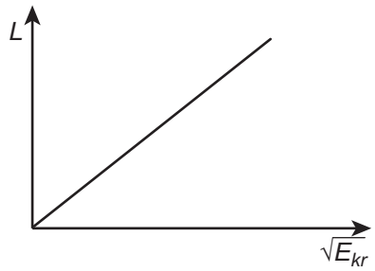
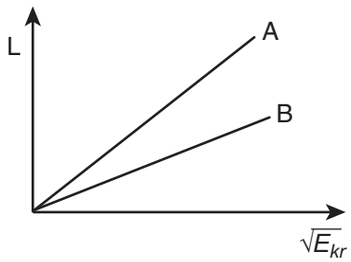
Graph of $\sqrt{E_{kr}}$ and L:



The shape of the graph is a straight line

ii) The slope of the graph gives the value of moment of inertia I.

iii) We know that the slope gives the value of moment of Inertia. The line A has higher slope and hence more moment of Inertia.



12. **Consider a thin uniform circular ring rolling down in an inclined plane without slipping. Compute the linear acceleration along the inclined plane if the angle of inclination is 45˚.**

Solution:

The linear acceleration along the inclined plane can be computed by

$$a = \frac{g\sin\theta}{1+\frac{K^2}{R^2}}$$

For a thin uniform circular ring, axis passing through its centre is $I = MR^2$

$$\therefore K^2 = R^2 \Rightarrow \frac{K^2}{R^2} = 1.$$

And the angle of inclination, θ = 45˚

$\Rightarrow (\sin 45^\circ = \frac{1}{\sqrt{2}})$

Hence,

$$a = \frac{\frac{g}{\sqrt{2}}}{1+1}$$

$$a = \frac{g}{2\sqrt{2}} ms^{-2}$$

COMPETITIVE EXAM CORNER

3BVAXL

Appendix-1.indd 289 28-01-2020 14:14:22